

# Jinu Lee

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## RESEARCH INTERESTS

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My research focuses on **structure-aware evaluation of reasoning traces**, aiming to understand and improve how natural language reasoning is done in LLMs.

- Understanding the structure of reasoning traces
- Evaluating and improving the reasoning capability of LLMs
- Natural language formalization for reasoning

## EDUCATION

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### University of Illinois Urbana-Champaign

Ph.D., Computer Science / Adviser: Dr. Julia Hockenmaier

Urbana-Champaign, IL

08/2024 - Present

### Seoul National University

B.S., Computer Science & Engineering

Seoul, Rep. of Korea

03/2018 - 08/2024

## WORK EXPERIENCE

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### Microsoft Research

Research Intern

Redmond, WA

05/2025 - 08/2025, 05/2026 - 08/2026

- Post-training LLM for program verification and theorem proving (2025)
- Autoformalizing tool-calling environment constraints and user intent (2026)

### LBOX

Research Engineer

Seoul, Rep. of Korea

07/2023 - 06/2024

- Neuro-symbolic reasoning; legal information extraction

### NCSOFT

Researcher

Seongnam, Rep. of Korea

07/2020 - 11/2020

- Korean dependency parsers and resources

## PUBLICATIONS

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### Conference proceedings.

[C7] **Jinu Lee**, Kyoung-woon On, Simeng Han, Arman Cohan, Julia Hockenmaier (2025). Evaluating Legal Reasoning Traces with Legal Issue Trees. *ACL 2026 Main Conference (Oral)*.

[C6] Seungone Kim, Ian Wu\*, **Jinu Lee\***, Xiang Yue, Seongyun Lee, Mingyeong Moon, Kiril Gashteovski, Caroline Lawrence, Julia Hockenmaier, Graham Neubig, Sean Welleck (2025). Scaling Evaluation-time Compute with Reasoning Models as Process Evaluators. *ACL 2026 Findings*.

[C5] **Jinu Lee**, Julia Hockenmaier (2025). Evaluating Step-by-step Reasoning Traces: A Survey. *EMNLP 2025 Findings*.

[C4] Chaeun Kim, **Jinu Lee**, Wonseok Hwang (2025). LegalSearchLM: Rethinking Legal Case Retrieval as Legal Elements Generation. *EMNLP 2025 Main Conference*.

[C3] **Jinu Lee**, Qi Liu, Runzhi Ma, Vincent Han, Ziqi Wang, Heng Ji, Julia Hockenmaier (2025). Entailment-Preserving First-Order Logic Representations for Natural Language Entailment. *ACL 2025 Main Conference*.

[C2] **Jinu Lee**, Wonseok Hwang (2025). SymBa: Symbolic Backward Chaining for Structured Natural Language Reasoning. *NAACL 2025 Main Conference*.

[C1] Youngwon Lee\*, **Jinu Lee\***, Seung-won Hwang (2023). Learning to Rank Generation with Pairwise Partial Rewards. *EMNLP 2023 Main Conference*.

## Preprints/Workshops.

[P3] Amruta Parulekar, **Jinu Lee**, Dilek Hakkani-Tur, Hari Sundaram (2025). Reasoning Consensus: Structural Ensembling of LLM Reasoning via Weighted DAG Aggregation. *arXiv preprint arXiv:2607.27783*.

[P2] **Jinu Lee**, Shivam Agarwal, Amruta Parulekar, Siddarth Madala, Dilek Hakkani-Tur, Julia Hockenmaier (2025). ReasoningFlow: Discourse Structures for Understanding LLM Reasoning Traces. *ArgMining workshop 2026 (non-archival)*.

[P1] Priyanka Kargupta\*, Stella Li\*, Haocheng Wang, **Jinu Lee**, Shan Chen, Orevaoghene Ahia, Dean Light, Thomas L. Griffiths, Max Kleiman-Weiner, Jiawei Han, Asli Celikyilmaz, Yulia Tsvetkov (2025). Cognitive Foundations for Reasoning and Their Manifestation in LLMs. *HC-AIR 2026 (non-archival)*.

\*: equal contribution.

## TEACHING, TALKS, AND SERVICE

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### Teaching

#### CS307 Modeling and Learning in Data Science

Teaching Assistant

Urbana, IL

01/2026-05/2026

#### CS440 Artificial Intelligence

Teaching Assistant

Urbana, IL

08/2025-12/2025

### Talks

#### ReasoningFlow: Semantic Structure of Complex Reasoning Traces

EleutherAI

Virtual

01/2026

#### Evaluating Legal Reasoning Traces with Legal Issue Trees

IDA.L Lab Seminar @ UoS

Seoul, Rep. of Korea

01/2026

#### Coupling Formal Logic and Natural Language Reasoning

Human-AI Collaborative Coding Lab @ Yonsei/KNU/UoS/SKKU

Seoul, Rep. of Korea

02/2025

### Academic Service

#### ACL Rolling Review (ARR) Reviewer

10/2024-Present

#### Conference on Language Modeling (CoLM) Reviewer

03/2025-Present

#### Midwest Speech & Language Day (MSLD) 2026 Student organizer

04/2026

#### NeurIPS Evaluation & Datasets Track Reviewer

05/2026-Present

## AWARDS AND HONOURS

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#### AICE Ph.D. Fellow

Awarded by Amazon-Illinois Center on AI for Interactive Conversational Experiences

08/2026 - 05/2027

#### KFAS Overseas Ph.D. Scholarship

Awarded by Korea Foundation of Advanced Studies

08/2024 - 05/2029

#### Presidential Science Scholarship for Undergraduates

Awarded by Korea Student Aid Foundation

03/2018 - 08/2023